



Section 1

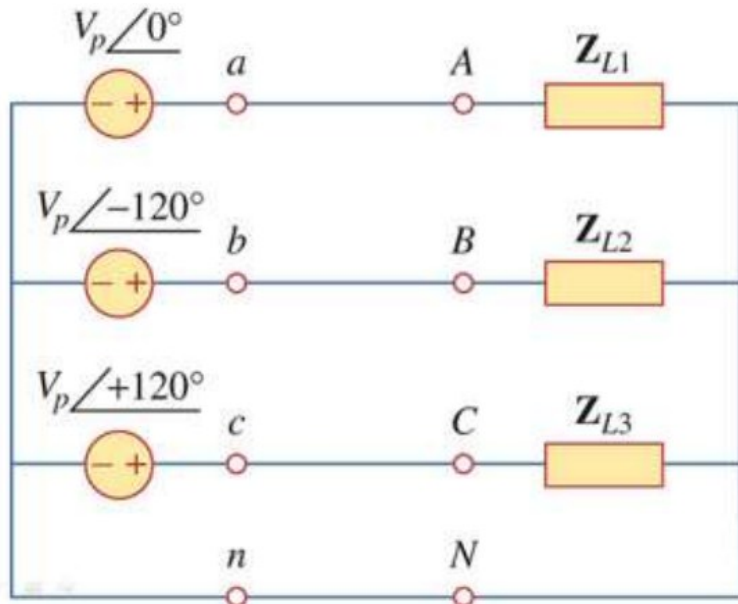
AC-CIRCUIT + MACHINERY PRINCIPLE

- **Three-phase circuit:**

Single-phase two-wire system: Single source connected to a load using two-wire system



Three sources connected to 3 loads using four-wire system Sources have EQUAL frequency but DIFFERENT phases.



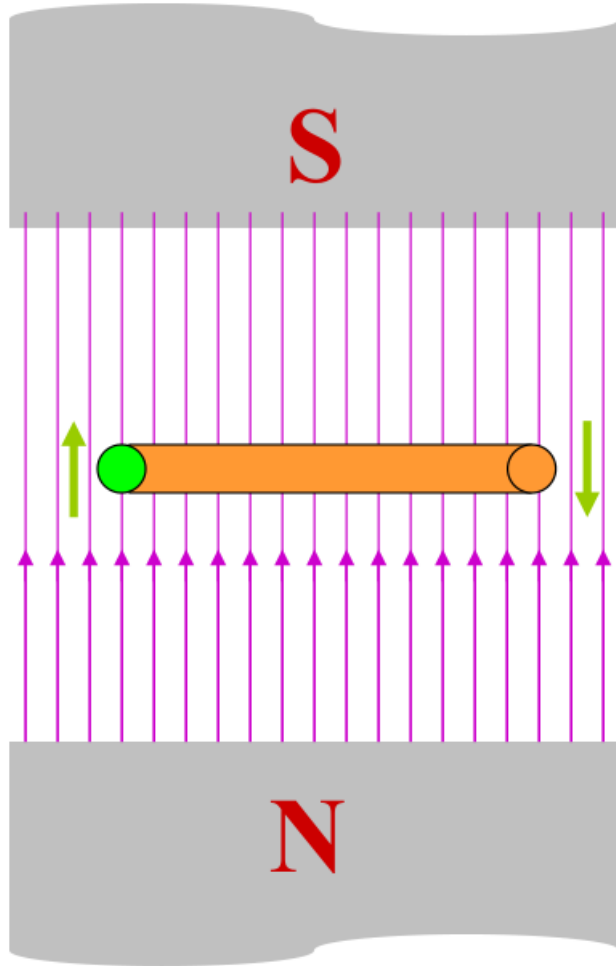
Three Phase System:

- A generator consists of three coils placed 120 degree apart.
- The voltage generated are equal in magnitude but, out of phase by 120 degree.
- Three phase is the most economical polyphase system.

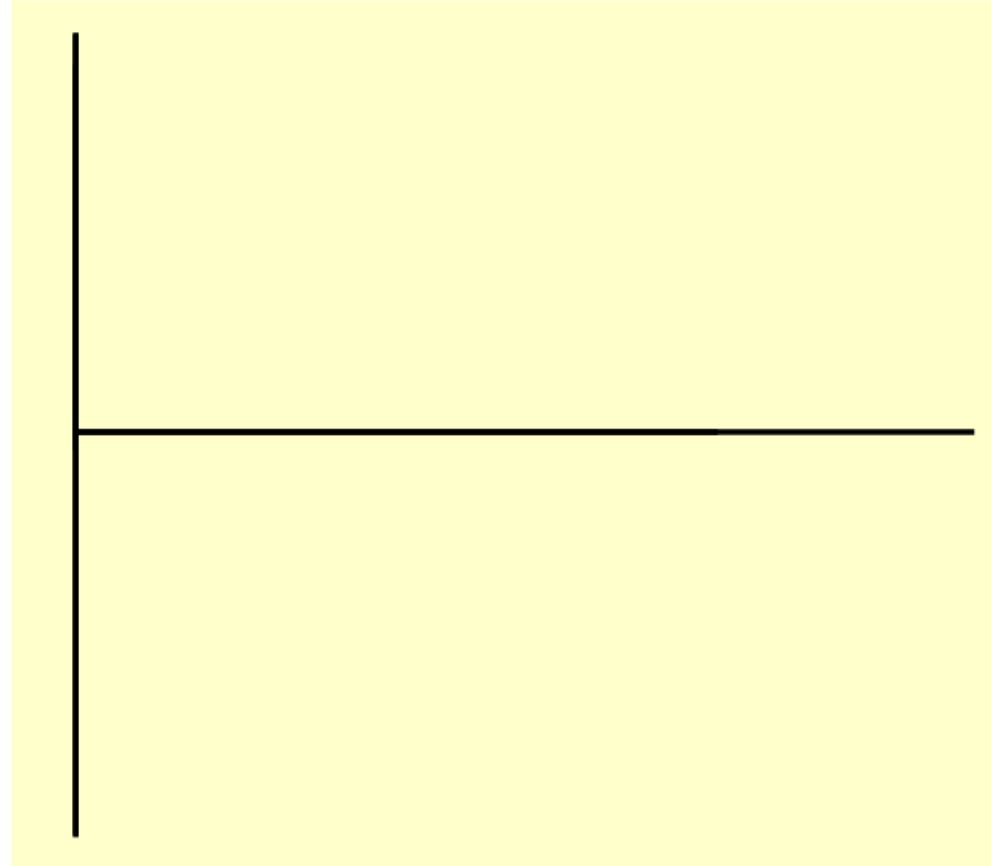
AC Generation

1. Three things must be present in order to produce electrical current:
 - a) Magnetic field
 - b) Conductor
 - c) Relative motion
2. Conductor cuts lines of magnetic flux, a voltage is induced in the conductor
3. Direction and speed are important

Generating a single phase



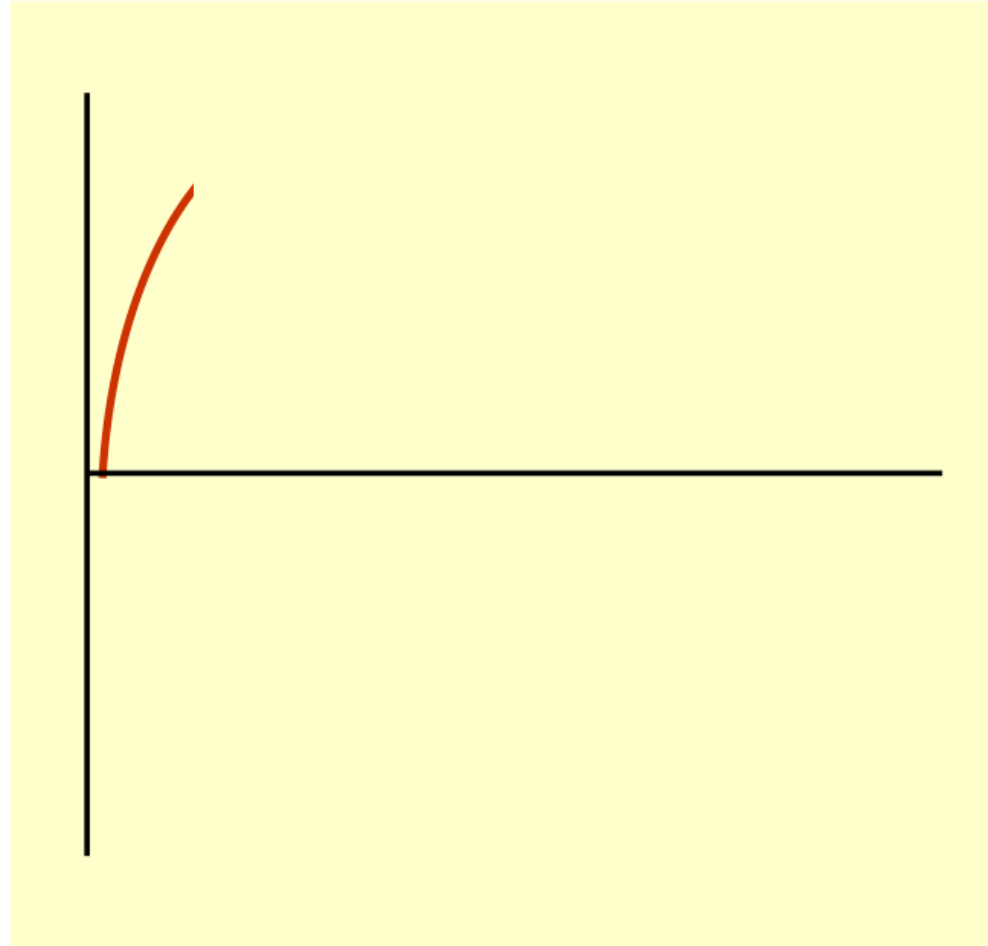
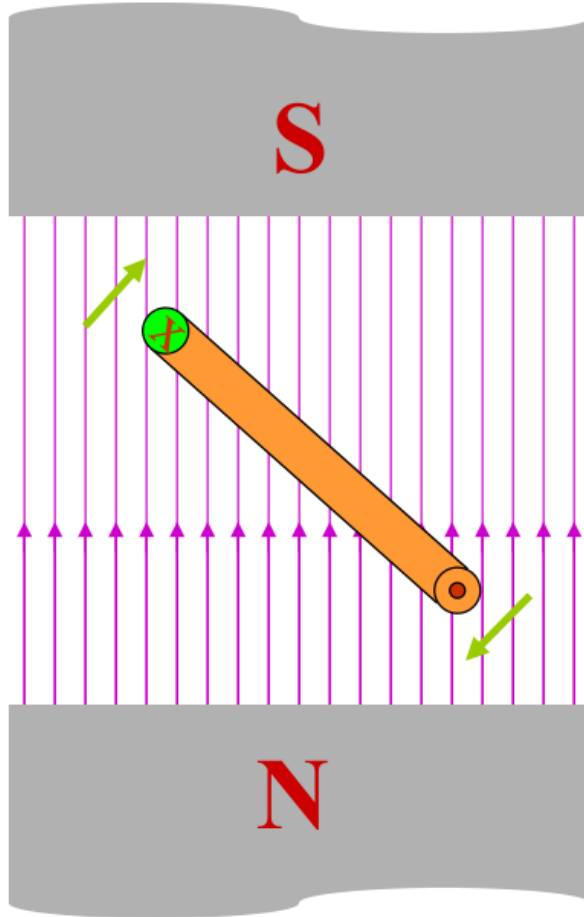
$$e(t) = N B A \omega \sin(\omega t)$$



Motion is parallel to the flux.

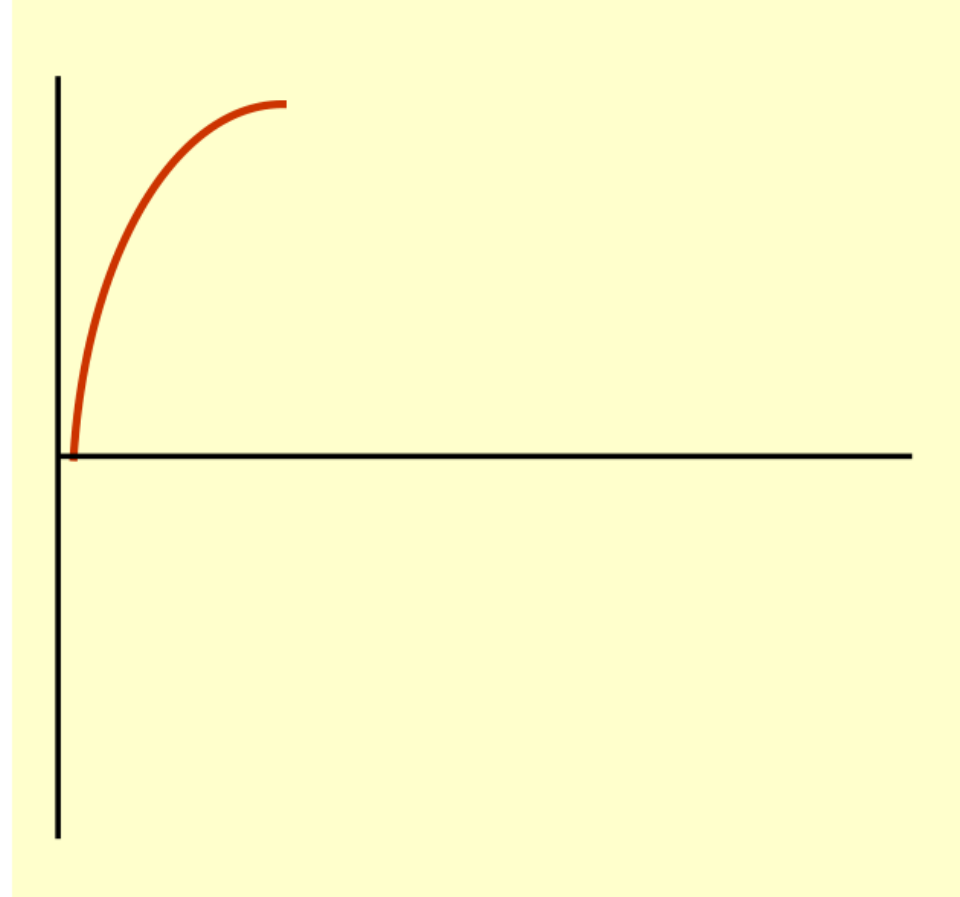
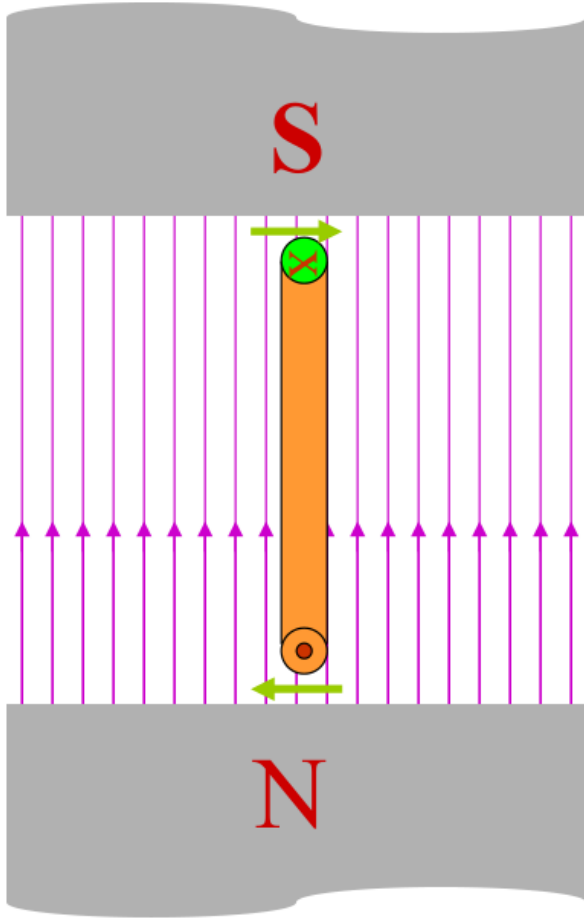
No voltage is induced.

Generating a single phase



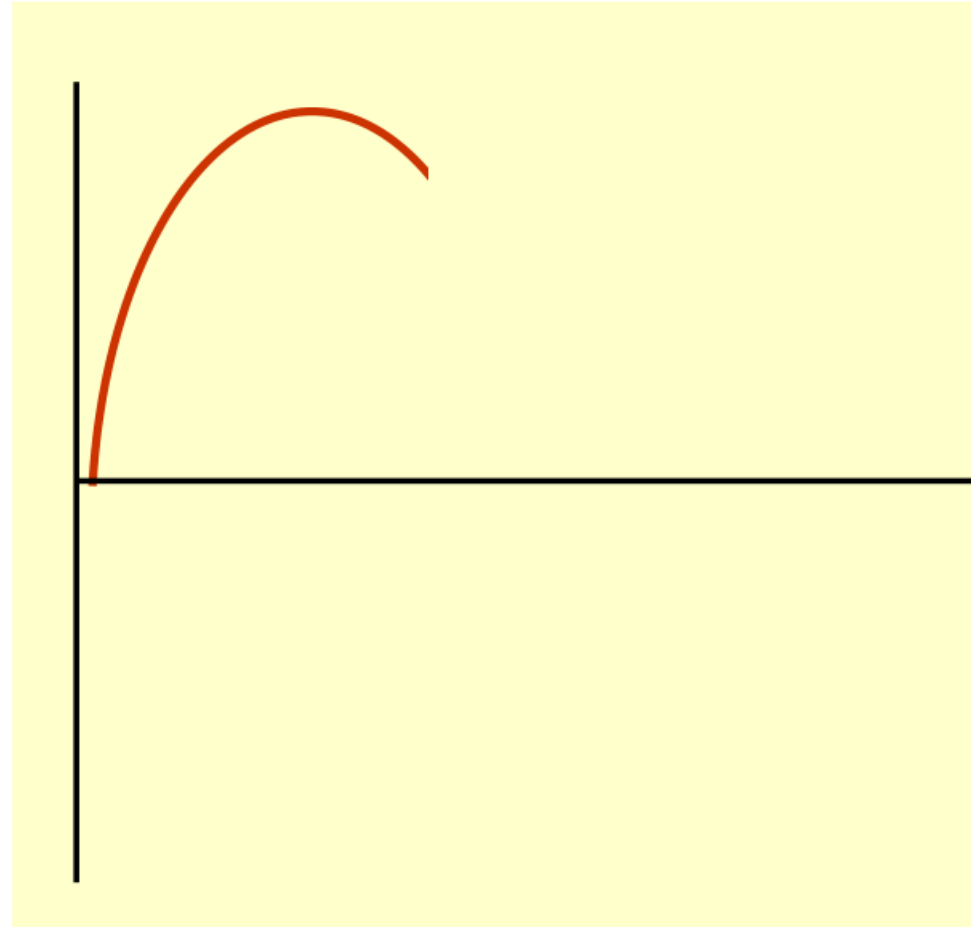
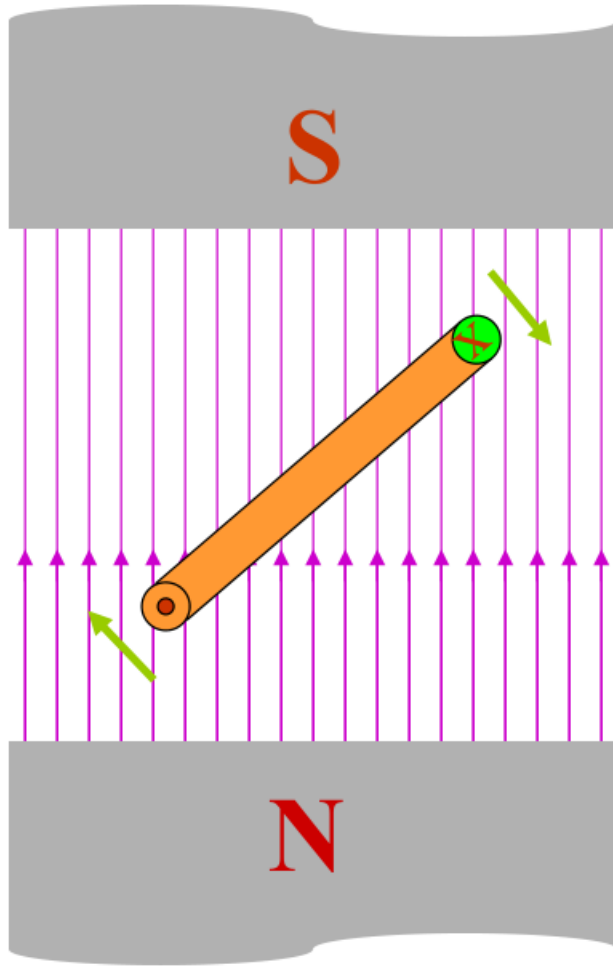
Motion is 45° to flux.
Induced voltage is 0.707 of maximum.

Generating a single phase



Motion is perpendicular to flux.
Induced voltage is maximum.

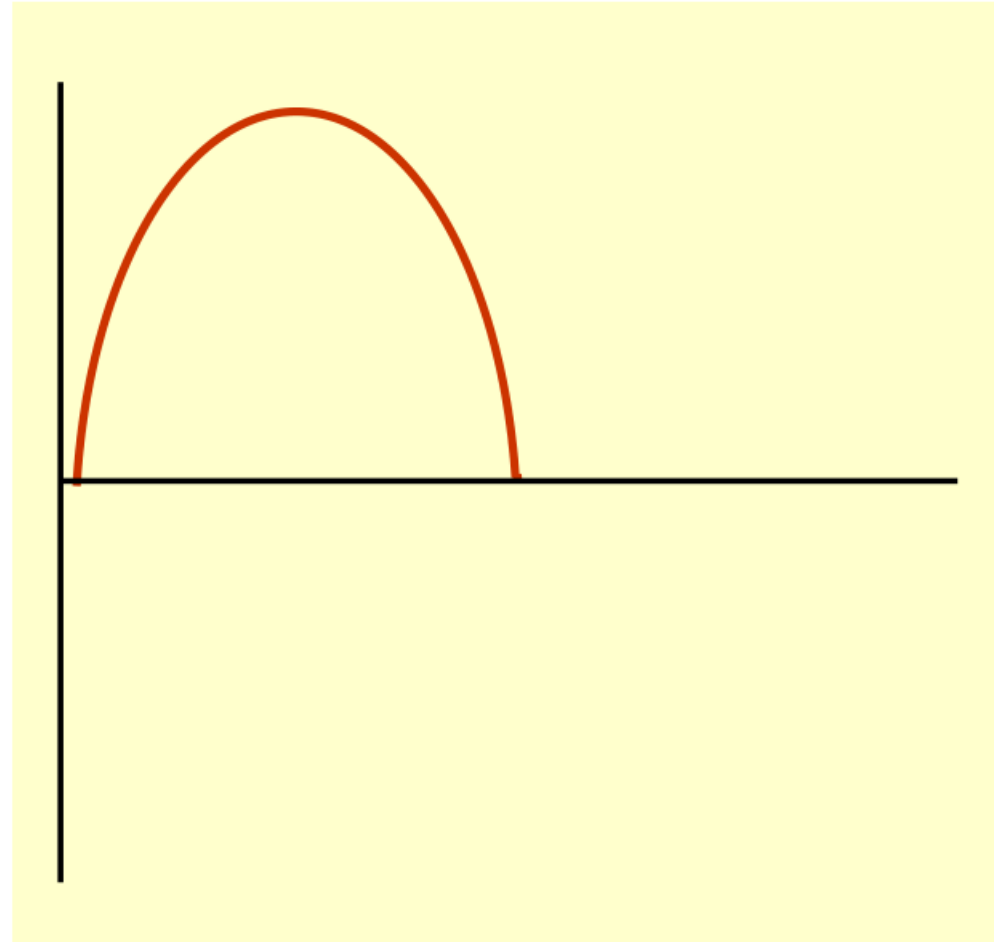
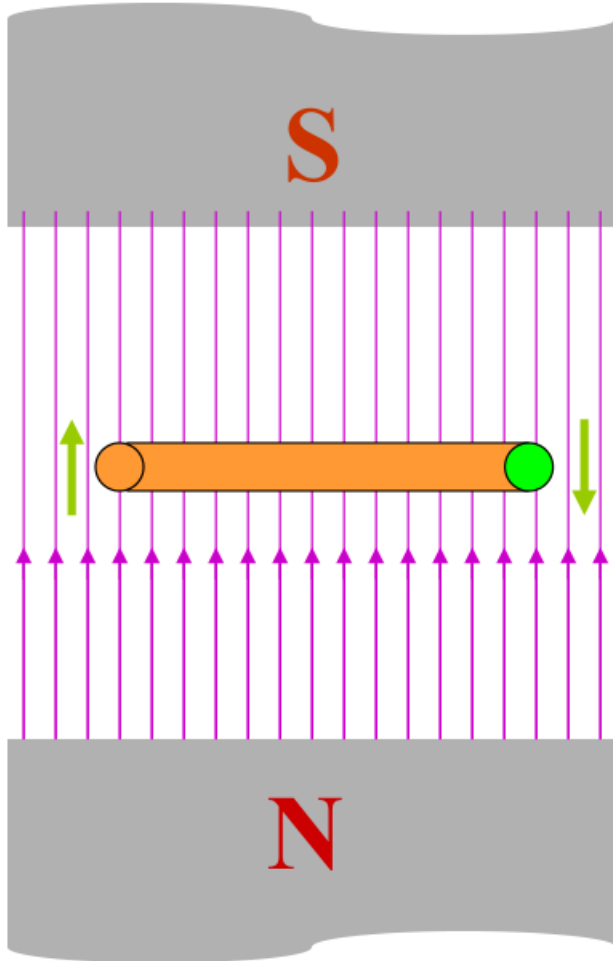
Generating a single phase



Motion is 45° to flux.

Induced voltage is 0.707 of maximum.

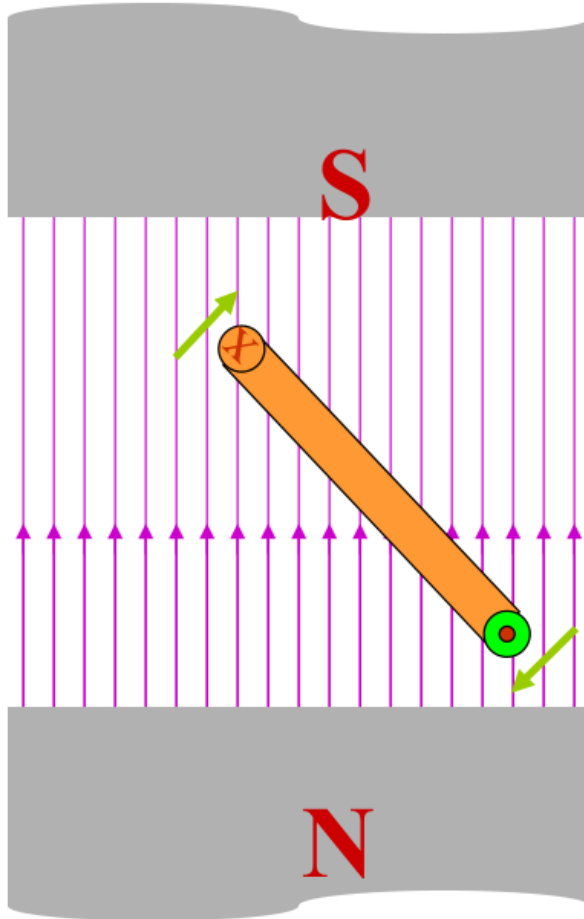
Generating a single phase



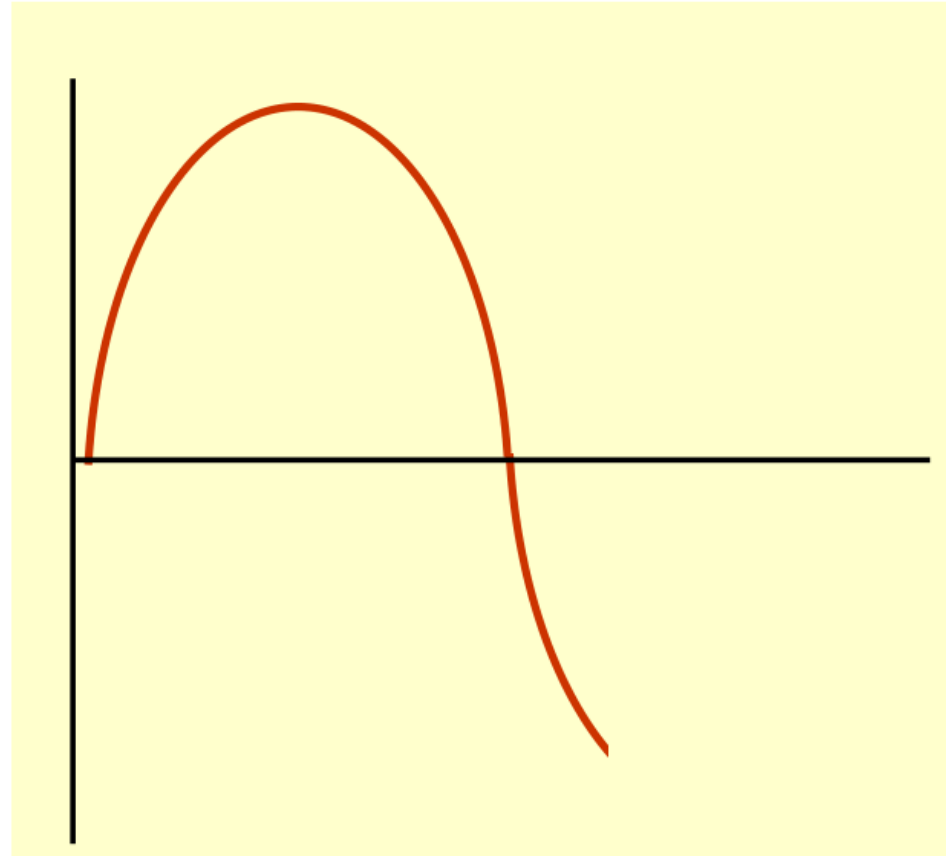
Motion is parallel to flux.

No voltage is induced.

Generating a single phase

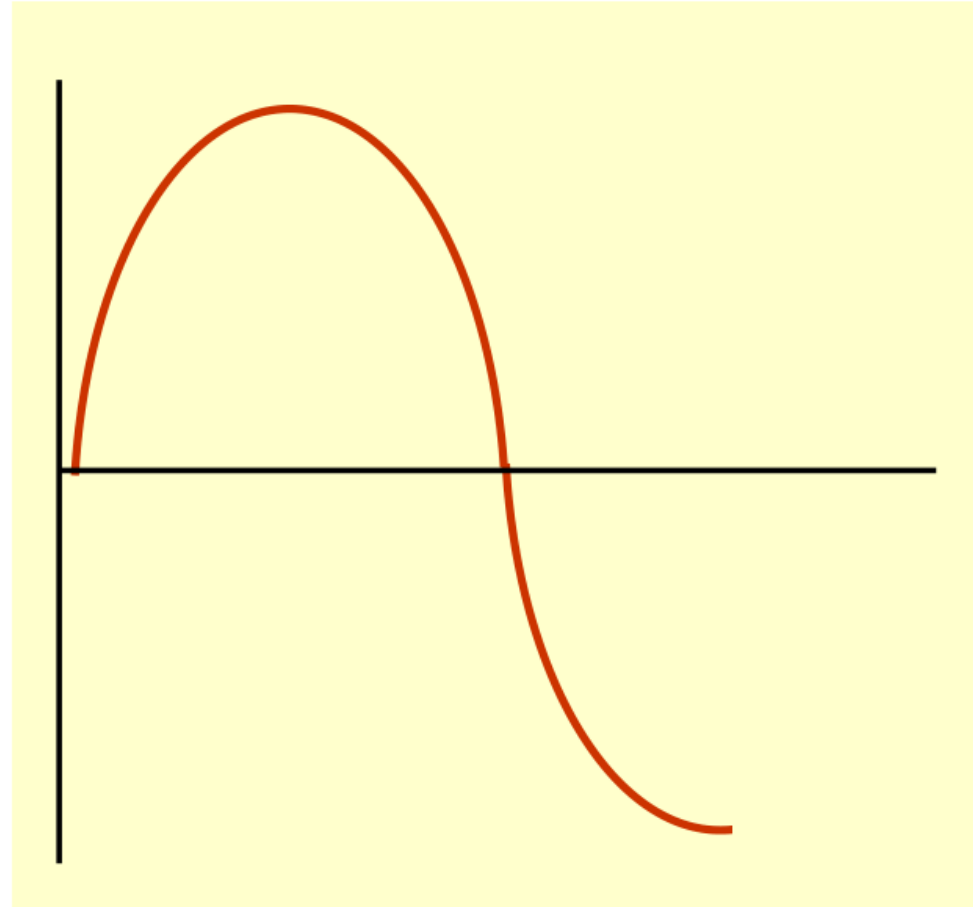
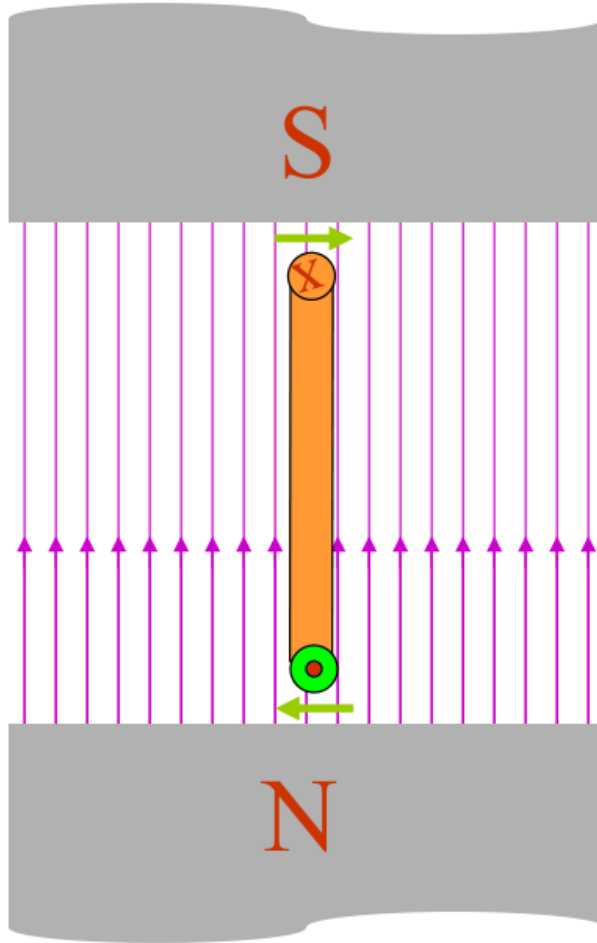


Notice current in the conductor has reversed.



Motion is 45° to flux.
Induced voltage is 0.707 of maximum.

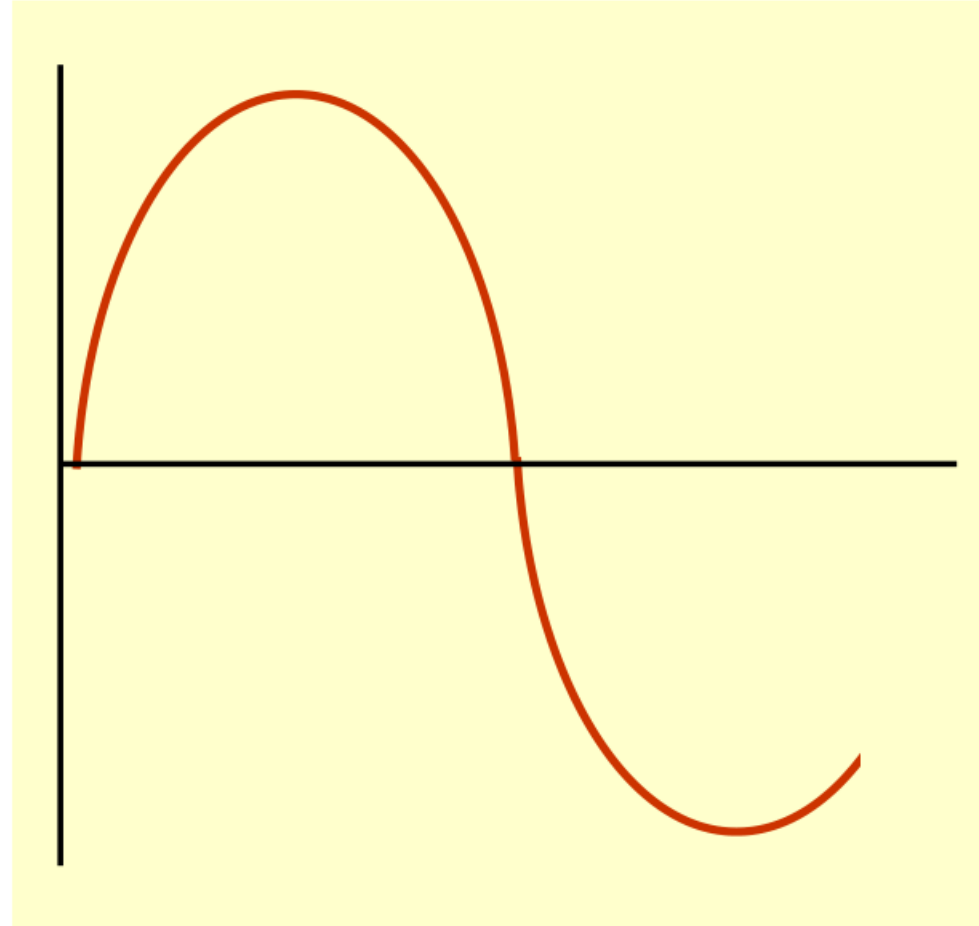
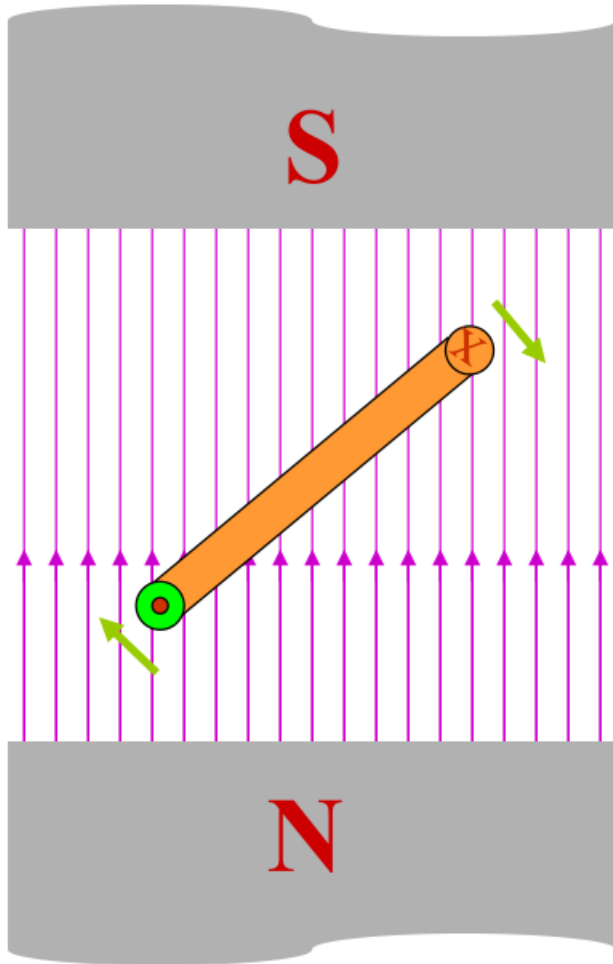
Generating a single phase



Motion is perpendicular to flux.

Induced voltage is maximum.

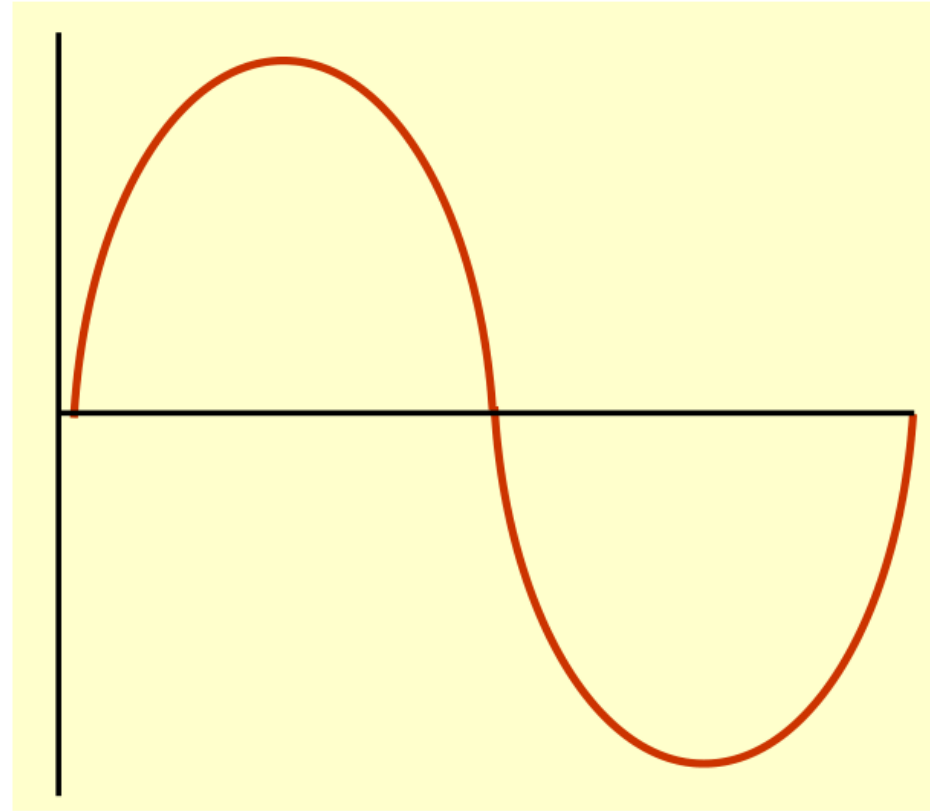
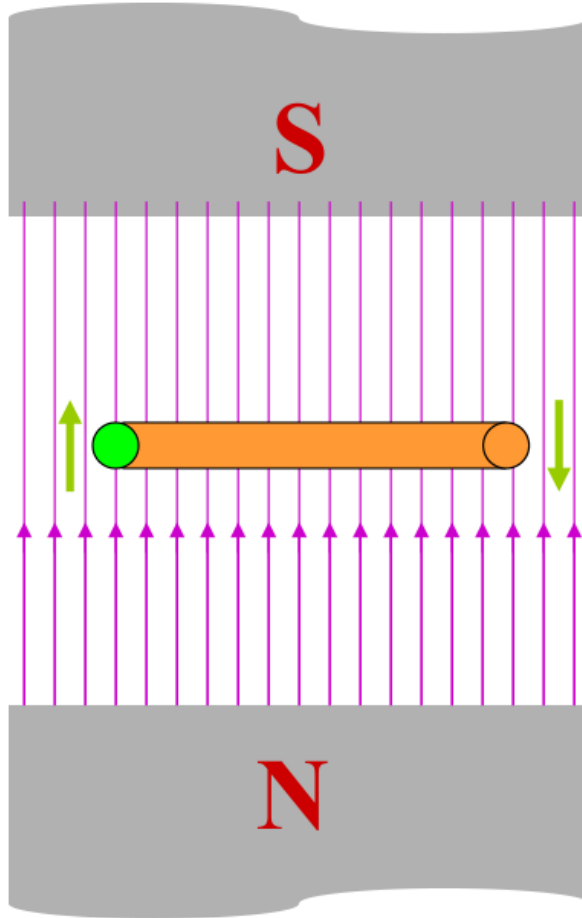
Generating a single phase



Motion is 45° to flux.

Induced voltage is 0.707 of maximum.

Generating a single phase



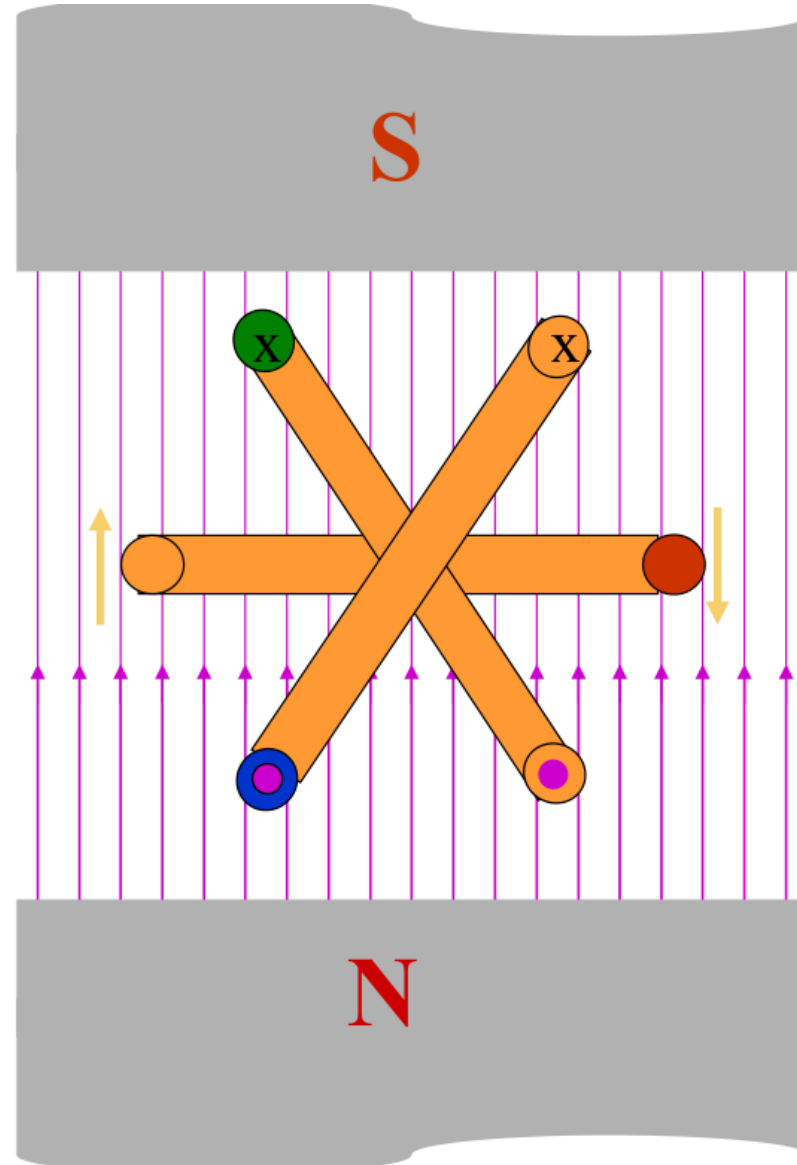
Motion is parallel to flux.

No voltage is induced.

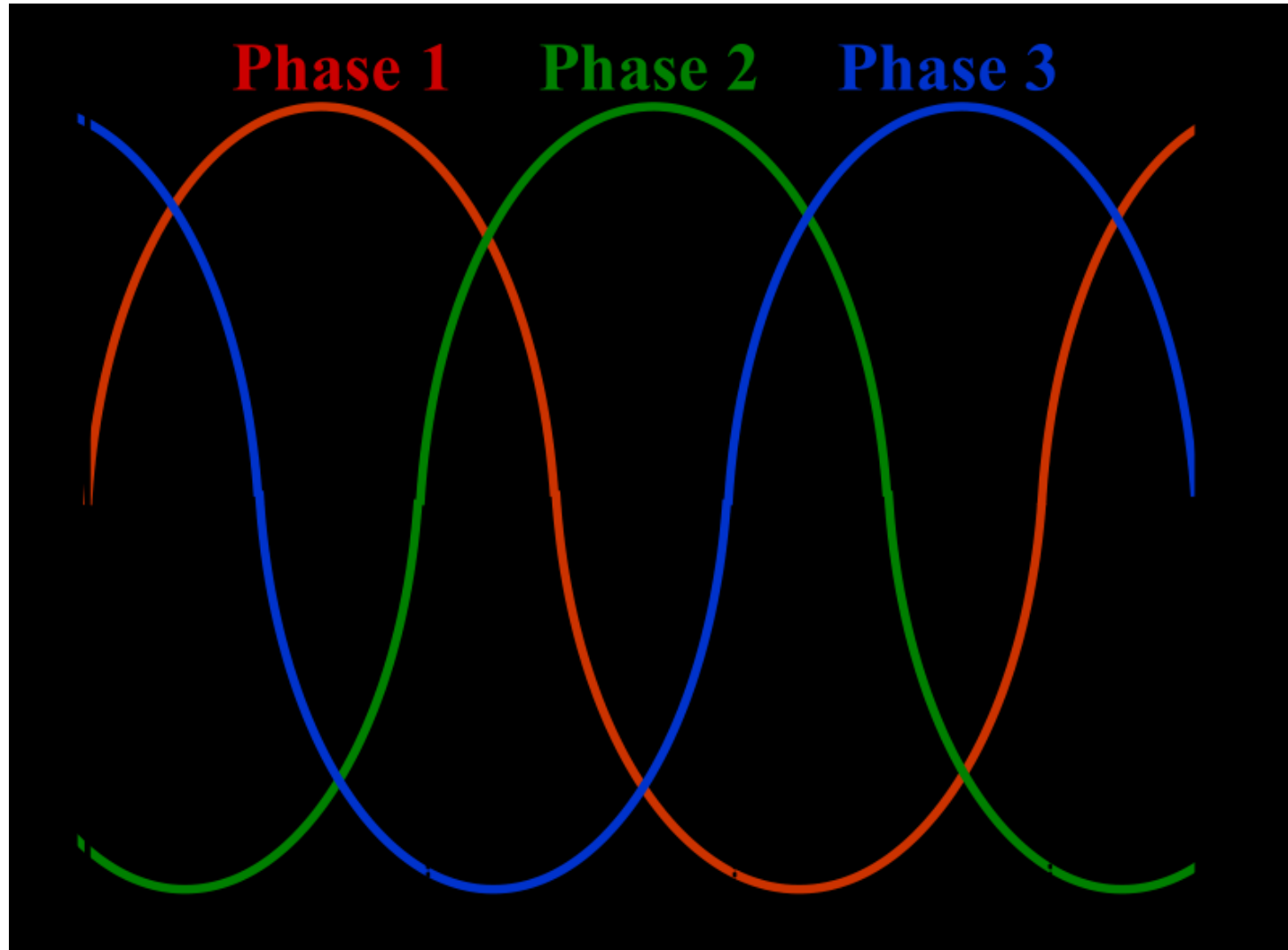
Ready to produce another cycle.

Generation of three-phase AC

Three Voltages will be induced across the coils with 120 phase difference

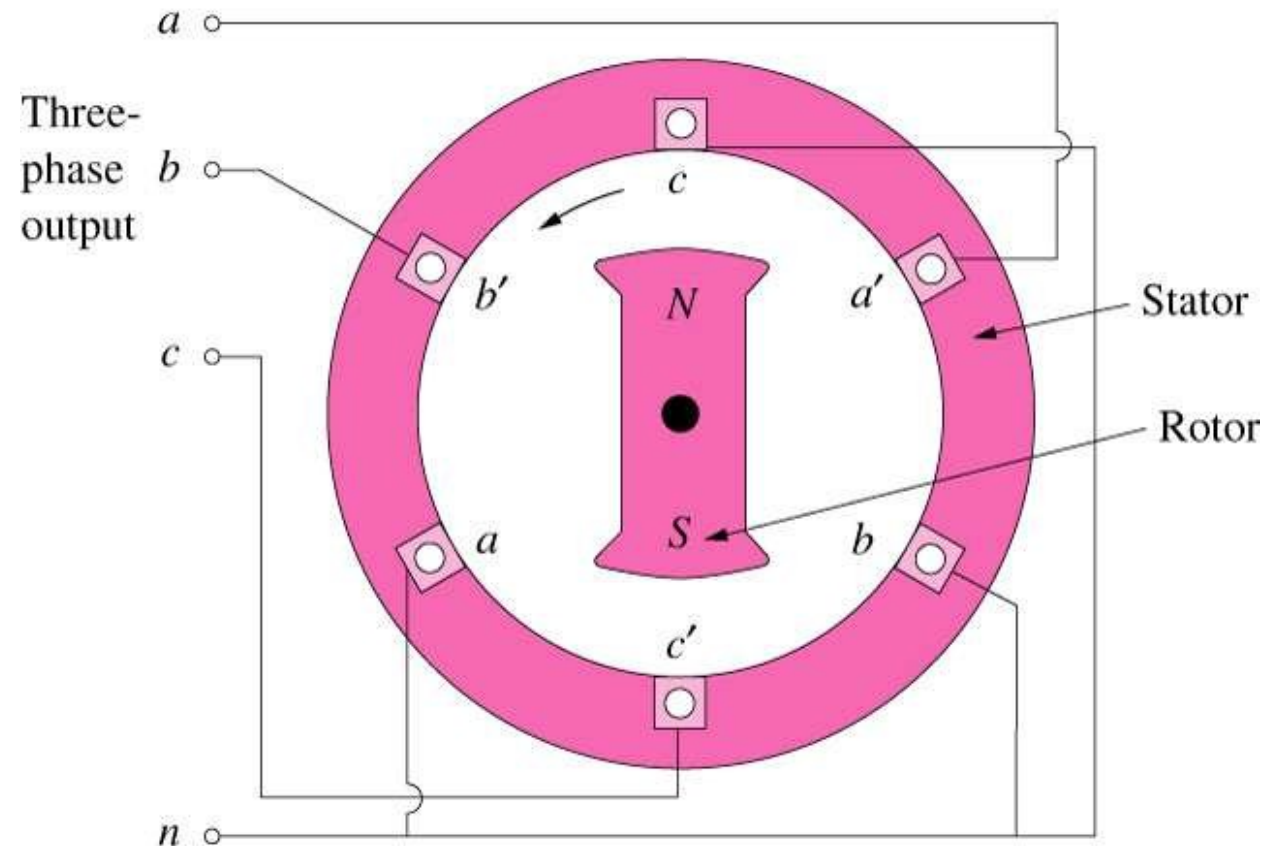


Three-phase Waveform

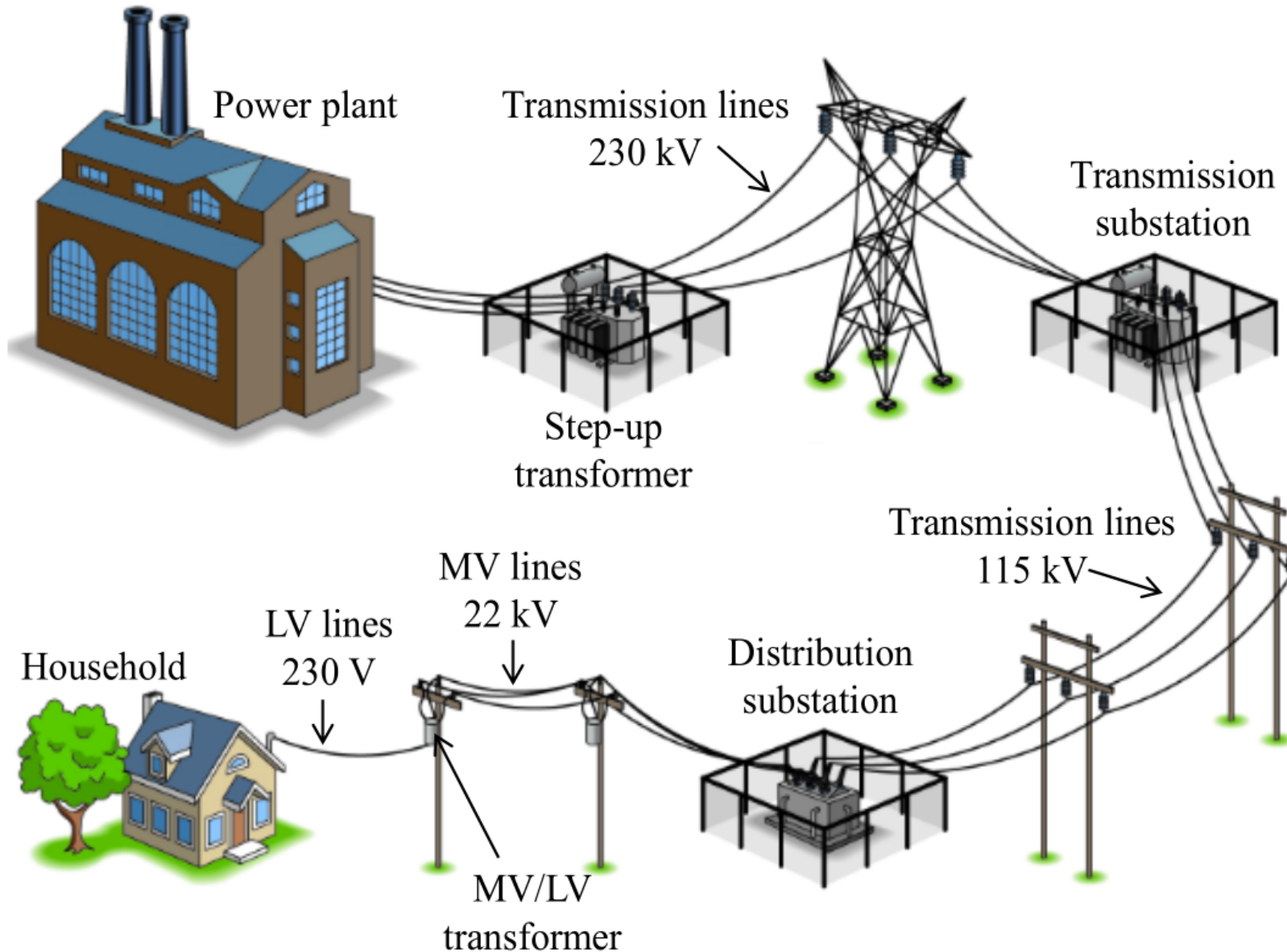


Practical Three Phase Generator

1. The generator consists of a rotating magnet (rotor) surrounded by a stationary winding (stator).
2. Three separate windings or coils with terminals a-a', b-b', and c-c' are physically placed 120 degree apart around the stator.
3. As the rotor rotates, its magnetic field cuts the flux from the three coils and induces voltages in the coils.
4. The induced voltage have equal magnitude but out of phase by 120 degree.



3-phase systems Generation, Transmission and Distribution



Y and Δ connections

Balanced 3-phase systems can be considered as 3 equal single phase voltage sources connected either as Y or Delta (Δ) to 3 single three loads connected as either Y or Δ .

SOURCE CONNECTIONS

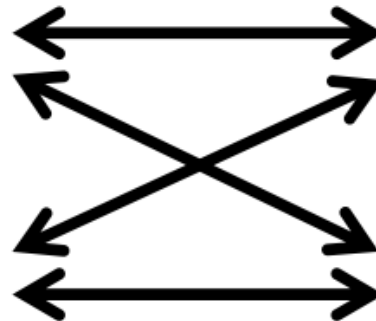
Y connected source

Δ connected source

LOAD CONNECTIONS

Y connected load

Δ connected load



Y-Y

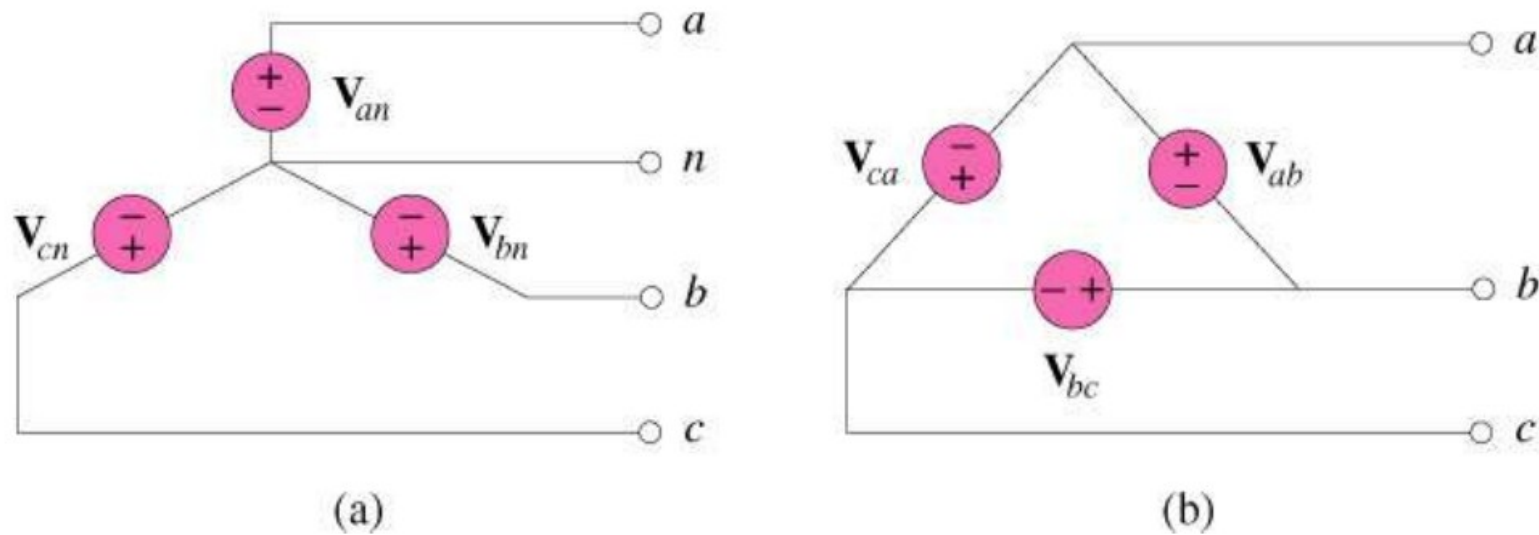
Y- Δ

Δ -Y

Δ - Δ

Balance Three-Phase Sources

Two possible configurations:

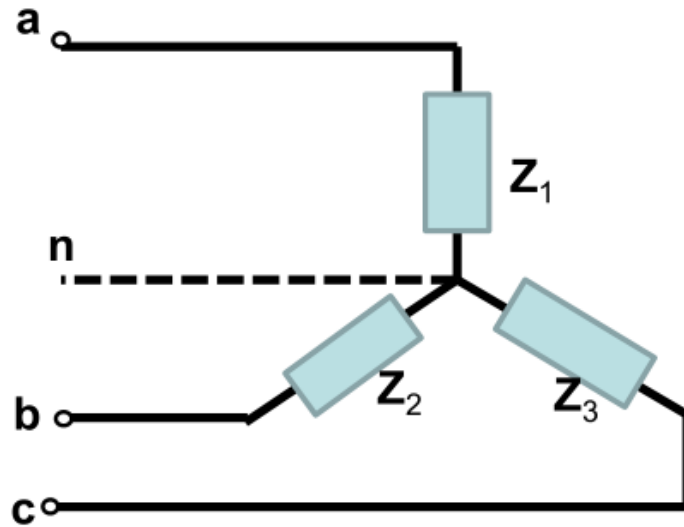


Three-phase voltage sources: (a) Y-connected ; (b) Δ -connected

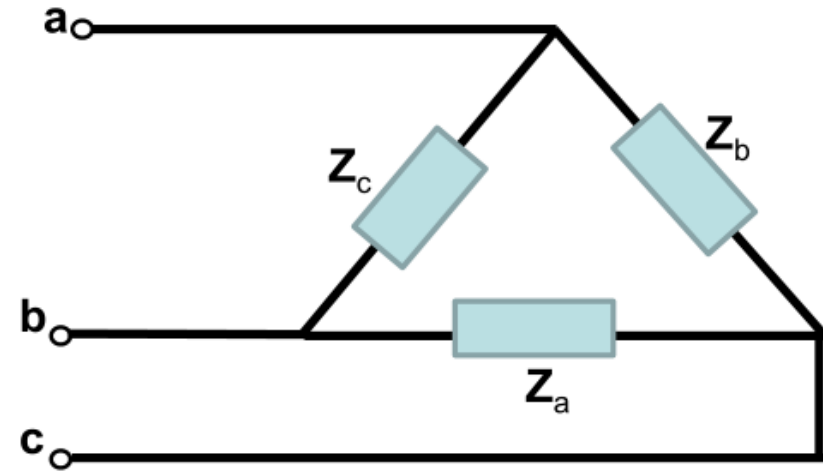
Y-connected network has a neutral node
 Δ -connected network has no neutral

Balance Three-Phase Loads

Y connection



Δ connection



Balanced load:

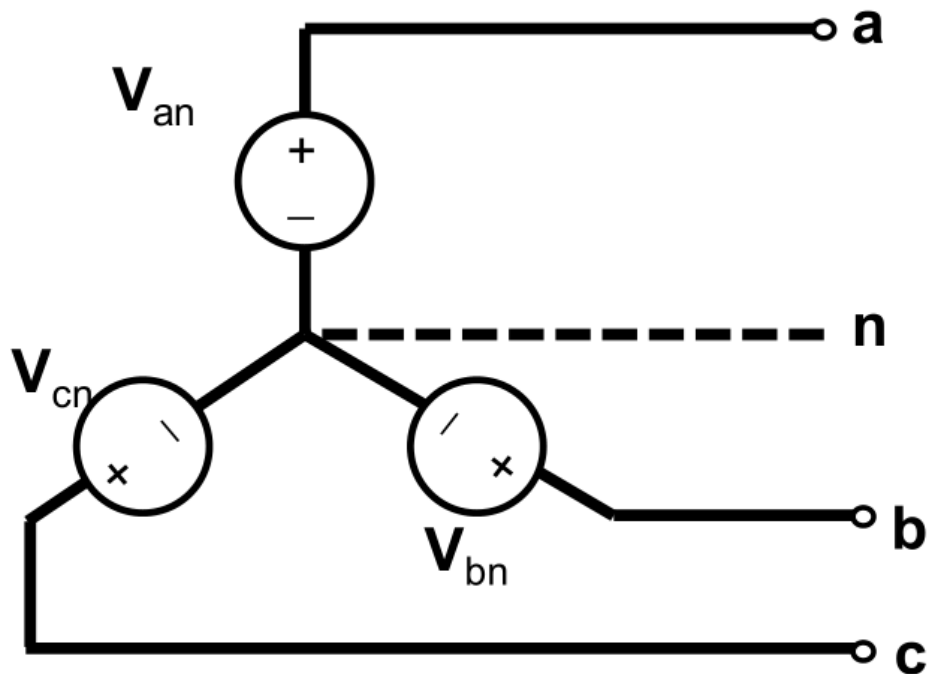
$$Z_1 = Z_2 = Z_3 = Z_Y$$

$$Z_a = Z_b = Z_c = Z_{\Delta}$$

Unbalanced load: each phase load may not be the same.

Phase Sequence

The phase sequence is the time order in which the voltages pass through their respective maximum values.



$$\Rightarrow V_{an} = V_p \angle 0^\circ$$

$$\Rightarrow V_{bn} = V_p \angle -120^\circ$$

$$\Rightarrow V_{cn} = V_p \angle 120^\circ$$

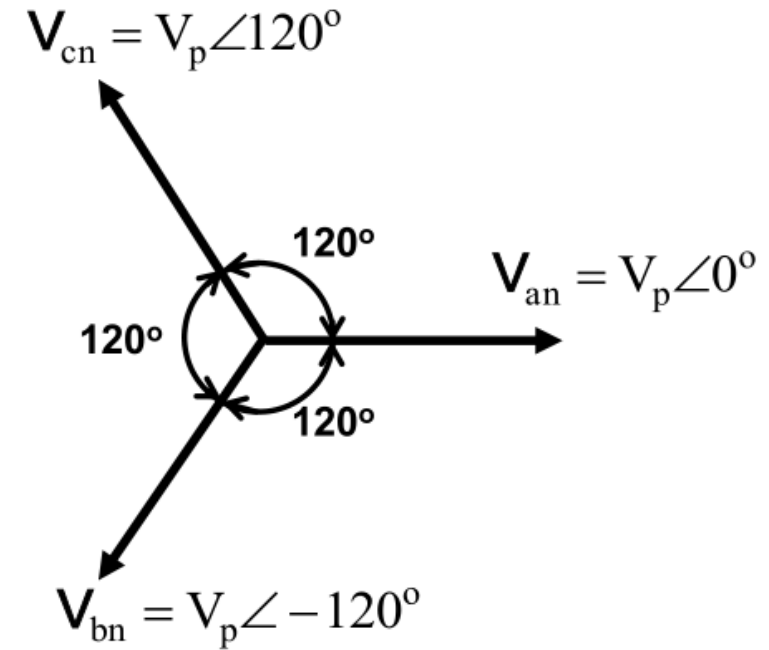
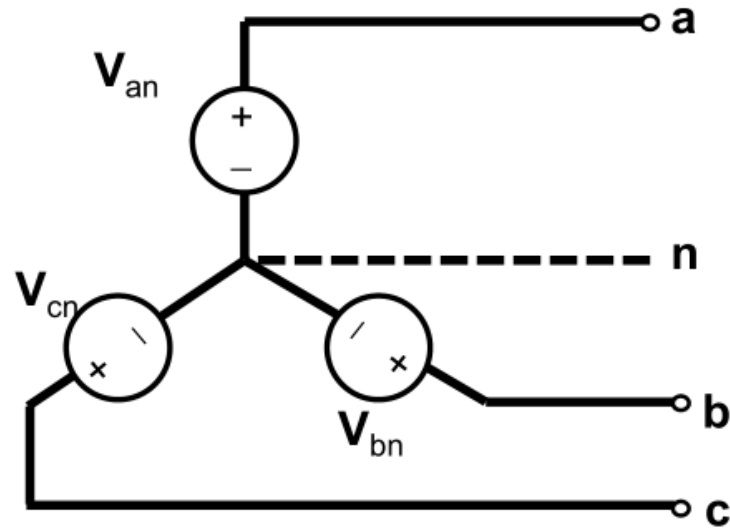
$$v_{an}(t) = \sqrt{2}V_p \cos(\omega t)$$

$$v_{bn}(t) = \sqrt{2}V_p \cos(\omega t - 120^\circ)$$

$$v_{cn}(t) = \sqrt{2}V_p \cos(\omega t + 120^\circ)$$

RMS phasors !

Phase Sequence

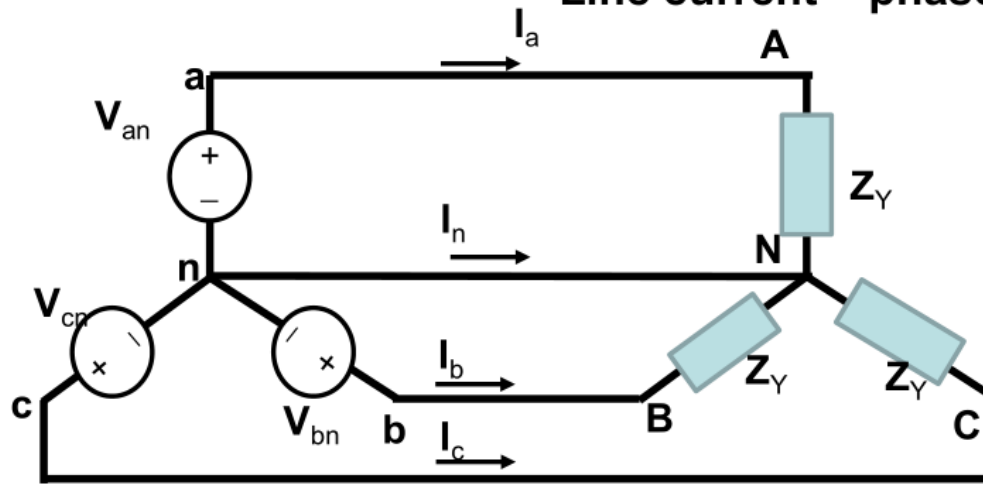


Phase sequence : V_{an} leads V_{bn} by 120° and V_{bn} leads V_{cn} by 120°

→ This is known as **abc sequence** or **positive sequence**

Balanced 3-phase Y-Y

Line current = phase current



$$V_{an} = V_p \angle 0^\circ$$

$$V_{bn} = V_p \angle -120^\circ$$

$$V_{cn} = V_p \angle 120^\circ$$

**Phase
voltages**

measured between the neutral
and any line
(line to neutral voltage)

$$I_a = \frac{V_p \angle 0^\circ}{Z_Y}$$

$$I_b = \frac{V_p \angle -120^\circ}{Z_Y}$$

$$I_c = \frac{V_p \angle 120^\circ}{Z_Y}$$

$$\therefore I_a + I_b + I_c = I_n = 0$$

**line
currents**

$$\begin{aligned} V_{ab} &= V_a - V_b = V_a - V_b + V_n - V_n = \\ &= V_{an} + V_{nb} = V_p \angle 0^\circ + V_p \angle 60^\circ \\ &= \sqrt{3} V_p \angle 30^\circ \end{aligned}$$

$$\begin{aligned} V_{bc} &= V_{bn} + V_{nc} \\ &= \sqrt{3} V_p \angle -90^\circ \end{aligned}$$

$$\begin{aligned} V_{ca} &= V_{cn} + V_{na} \\ &= \sqrt{3} V_p \angle 150^\circ \end{aligned}$$

**line-line
voltages
OR
Line
voltages**

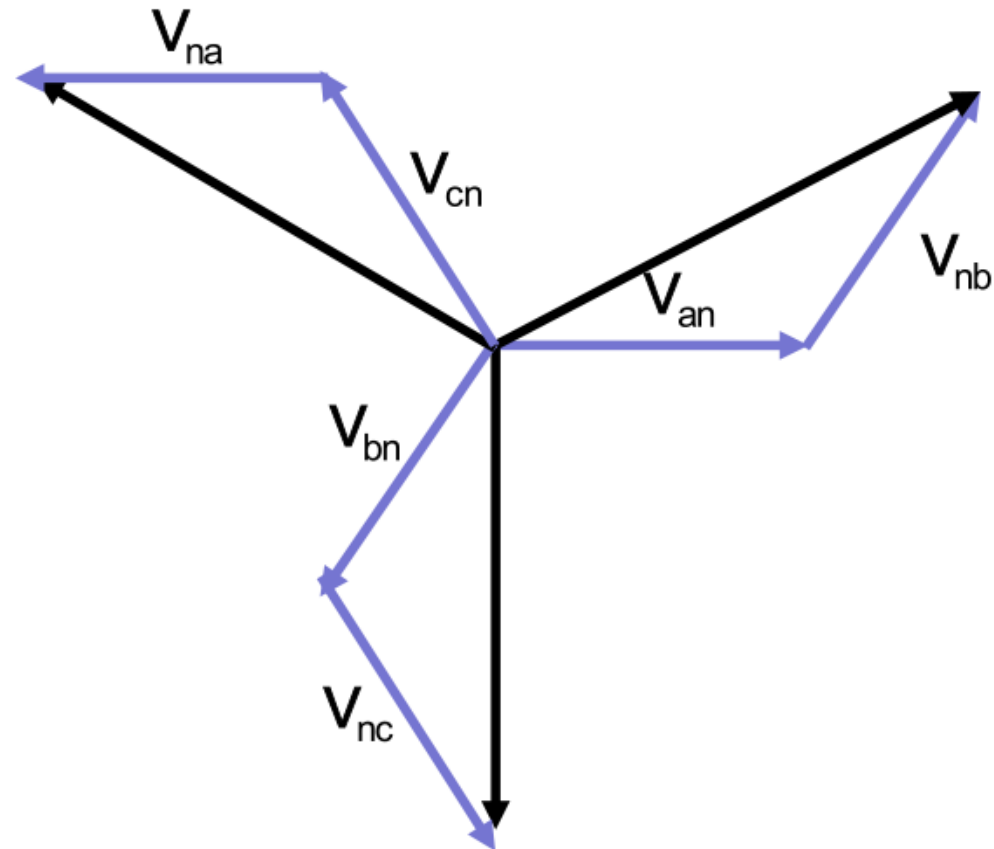
The wire connecting n and N can be removed !

Balanced 3-phase Y-Y

$$\begin{aligned}V_{ab} &= V_{an} + V_{nb} \\&= V_p \angle 0^\circ + V_p \angle 60^\circ \\&= \sqrt{3}V_p \angle 30^\circ\end{aligned}$$

$$\begin{aligned}V_{bc} &= V_{bn} + V_{nc} \\&= \sqrt{3}V_p \angle -90^\circ\end{aligned}$$

$$\begin{aligned}V_{ca} &= V_{cn} + V_{na} \\&= \sqrt{3}V_p \angle 150^\circ\end{aligned}$$

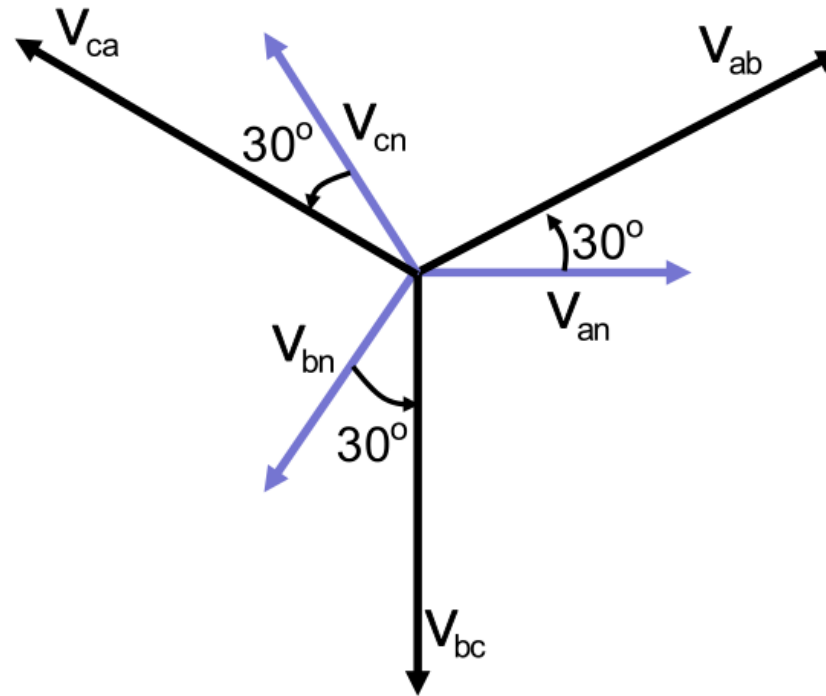


Balanced 3-phase Y-Y

$$\begin{aligned}V_{ab} &= V_{an} + V_{nb} \\&= V_p \angle 0^\circ + V_p \angle 60^\circ \\&= \sqrt{3}V_p \angle 30^\circ\end{aligned}$$

$$\begin{aligned}V_{bc} &= V_{bn} + V_{nc} \\&= \sqrt{3}V_p \angle -90^\circ\end{aligned}$$

$$\begin{aligned}V_{ca} &= V_{cn} + V_{na} \\&= \sqrt{3}V_p \angle 150^\circ\end{aligned}$$



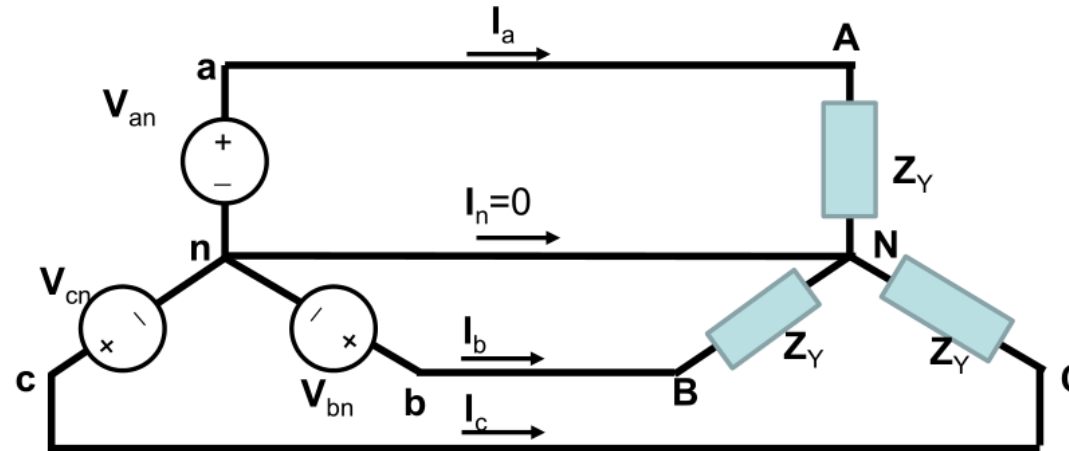
$$V_L = \sqrt{3}V_p$$

where $V_L = |V_{ab}| = |V_{bc}| = |V_{ca}|$ and $V_p = |V_{an}| = |V_{bn}| = |V_{cn}|$

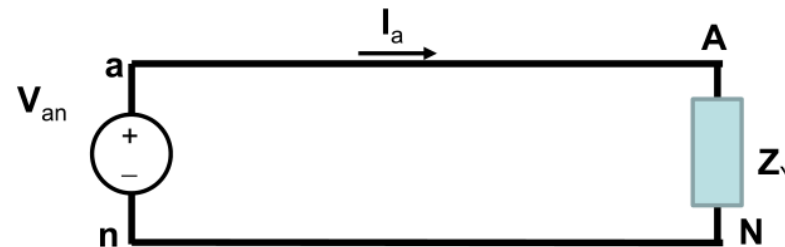
Line voltage LEADS phase voltage by 30°

Balanced 3-phase Y-Y

For a **balanced Y-Y** connection, analysis can be performed using an **equivalent per-phase** circuit: e.g. for phase A:



For a **balanced Y-Y** connection, analysis can be performed using an **equivalent per-phase** circuit: e.g. for phase A:

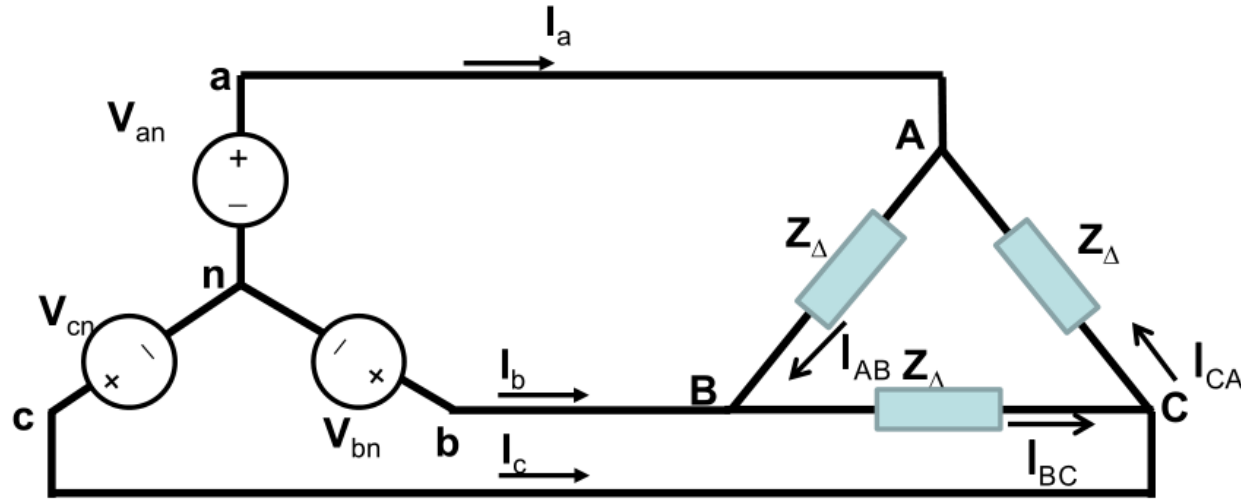


$$I_a = \frac{V_{an}}{Z_Y}$$

$$I_b = I_a \angle -120^\circ$$

$$I_c = I_a \angle 120^\circ$$

Balanced 3-phase Y-Δ



$$V_{an} = V_p \angle 0^\circ$$

$$V_{bn} = V_p \angle -120^\circ$$

$$V_{cn} = V_p \angle 120^\circ$$

$$\begin{aligned} V_{ab} &= \sqrt{3}V_p \angle 30^\circ \\ &= V_{AB} \end{aligned}$$

$$\begin{aligned} V_{bc} &= \sqrt{3}V_p \angle -90^\circ \\ &= V_{BC} \end{aligned}$$

$$\begin{aligned} V_{ca} &= \sqrt{3}V_p \angle 150^\circ \\ &= V_{CA} \end{aligned}$$

$$I_{AB} = \frac{V_{AB}}{Z_{\Delta}}$$

$$I_{BC} = \frac{V_{BC}}{Z_{\Delta}}$$

$$I_{CA} = \frac{V_{CA}}{Z_{\Delta}}$$

Using KCL, $I_a = I_{AB} - I_{CA}$

$$\begin{aligned} &= I_{AB}(1 - 1 \angle 120^\circ) \\ &= I_{AB} \sqrt{3} \angle -30^\circ \end{aligned}$$

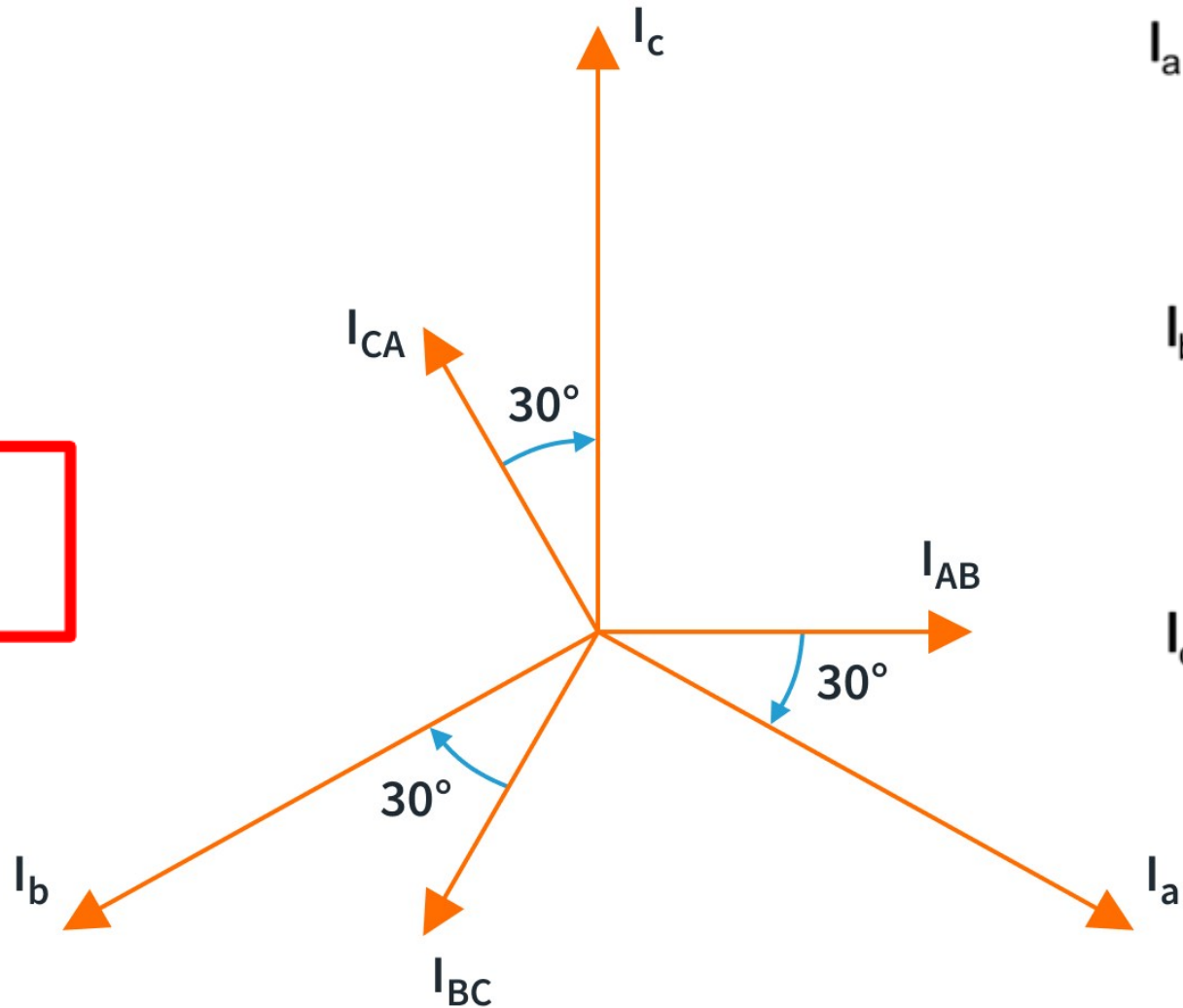
$$\begin{aligned} I_b &= I_{BC} - I_{AB} \\ &= I_{BC}(1 - 1 \angle 120^\circ) \\ &= I_{BC} \sqrt{3} \angle -30^\circ \end{aligned}$$

$$I_c = I_{CA} \sqrt{3} \angle -30^\circ$$

Phase currents

Balanced 3-phase Y-Δ

$$I_L = \sqrt{3}I_p$$

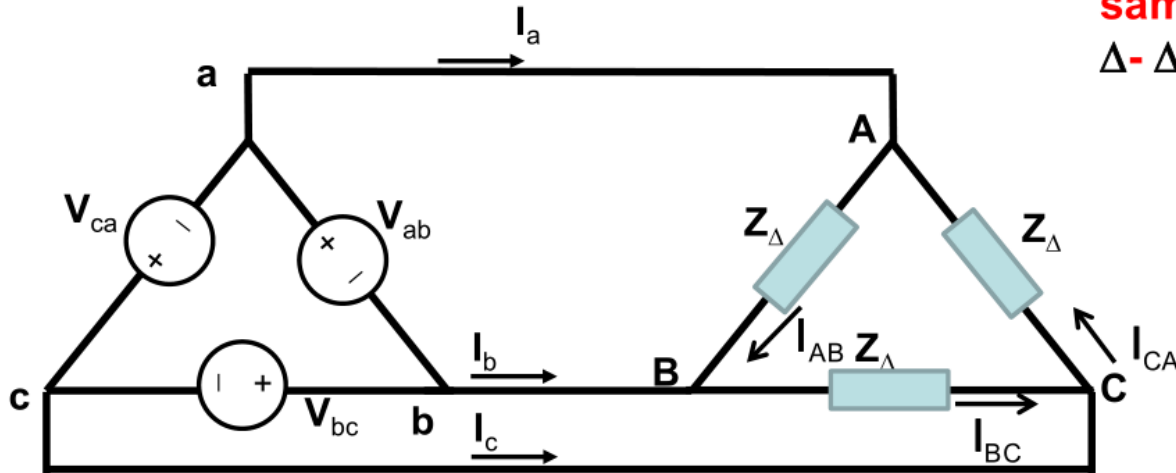


$$\begin{aligned} I_a &= I_{AB} - I_{CA} \\ &= I_{AB}(1 - 1\angle 120^\circ) \\ &= I_{AB}\sqrt{3}\angle -30^\circ \end{aligned}$$

$$\begin{aligned} I_b &= I_{BC} - I_{AB} \\ &= I_{BC}(1 - 1\angle 120^\circ) \\ &= I_{BC}\sqrt{3}\angle -30^\circ \end{aligned}$$

$$I_c = I_{CA}\sqrt{3}\angle -30^\circ$$

Balanced 3-phase Δ - Δ



Line-line voltage is the same as phase voltage in

Δ - Δ

$$V_{ab} = V_p \angle 0^\circ$$

$$V_{bc} = V_p \angle -120^\circ$$

$$V_{ca} = V_p \angle 120^\circ$$

$$V_{ab} = V_{AB}$$

$$V_{bc} = V_{BC}$$

$$V_{ca} = V_{CA}$$

$$I_{AB} = \frac{V_{AB}}{Z_\Delta}$$

$$I_{BC} = \frac{V_{BC}}{Z_\Delta}$$

$$I_{CA} = \frac{V_{CA}}{Z_\Delta}$$

Using KCL, $I_a = I_{AB} - I_{CA}$

$$= I_{AB}(1 - 1 \angle 120^\circ)$$

$$= I_{AB} \sqrt{3} \angle -30^\circ$$

$$I_b = I_{BC} - I_{AB}$$

$$= I_{BC}(1 - 1 \angle 120^\circ)$$

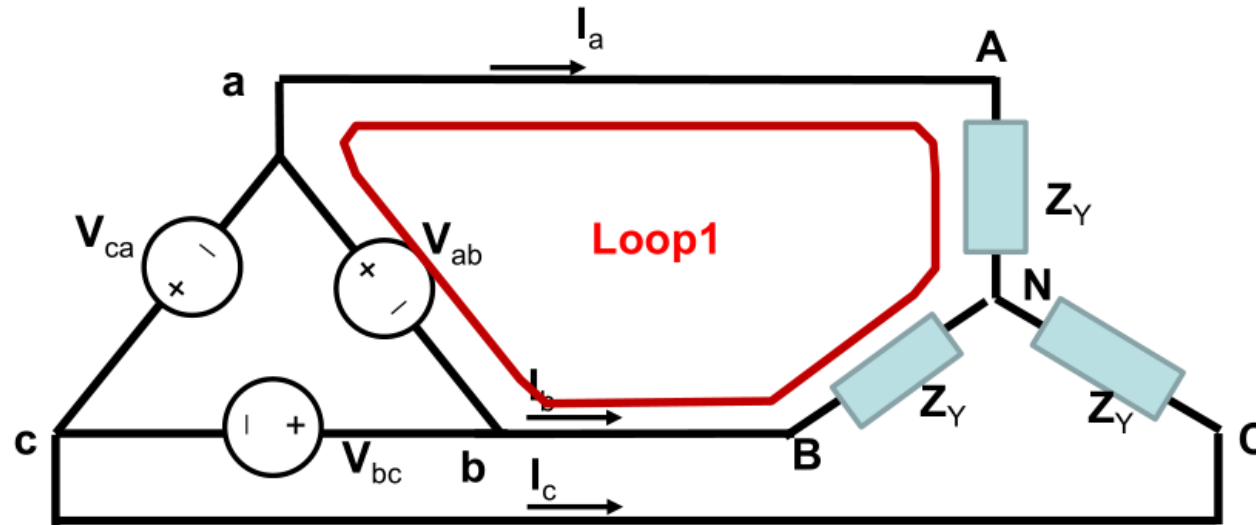
$$= I_{BC} \sqrt{3} \angle -30^\circ$$

$$I_c = I_{CA} \sqrt{3} \angle -30^\circ$$

Phase currents

line current

Balanced 3-phase Δ -Y



$$V_{ab} = V_p \angle 0^\circ$$

$$V_{bc} = V_p \angle -120^\circ$$

$$V_{ca} = V_p \angle 120^\circ$$

How to find I_a ?

$$\text{Loop1} \quad -V_{ab} + Z_Y I_a - Z_Y I_b = 0 \quad \Rightarrow I_a - I_b = \frac{V_{ab}}{Z_Y}$$

$$\text{Since circuit is balanced, } I_b = I_a \angle -120^\circ \quad \Rightarrow I_a - I_b = I_a (1 - 1 \angle -120^\circ)$$

$$= I_a \sqrt{3} \angle 30^\circ$$

$$\text{Therefore } I_a = \frac{V_p / \sqrt{3}}{Z_Y} \angle -30^\circ$$

Example I:

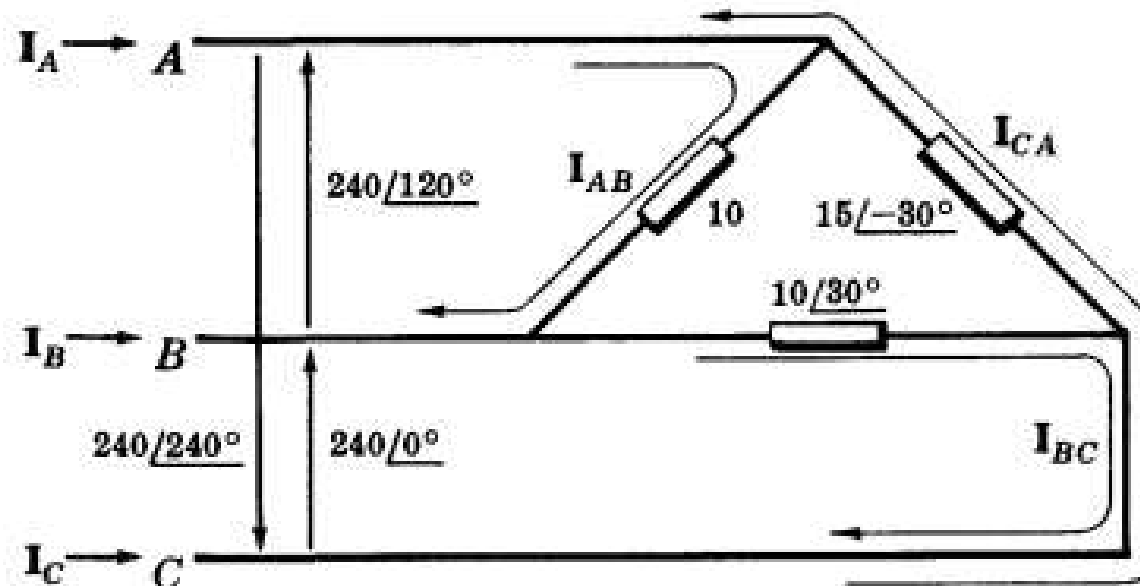
A three-phase, three-wire, 240V, ABC system has a delta-connected load with

$$Z_{AB} = 10 \angle 0^\circ$$

$$Z_{BC} = 10 \angle 30^\circ$$

$$Z_{CA} = 15 \angle -30^\circ$$

Obtain the line currents and draw the phasor diagram?



Example 2:

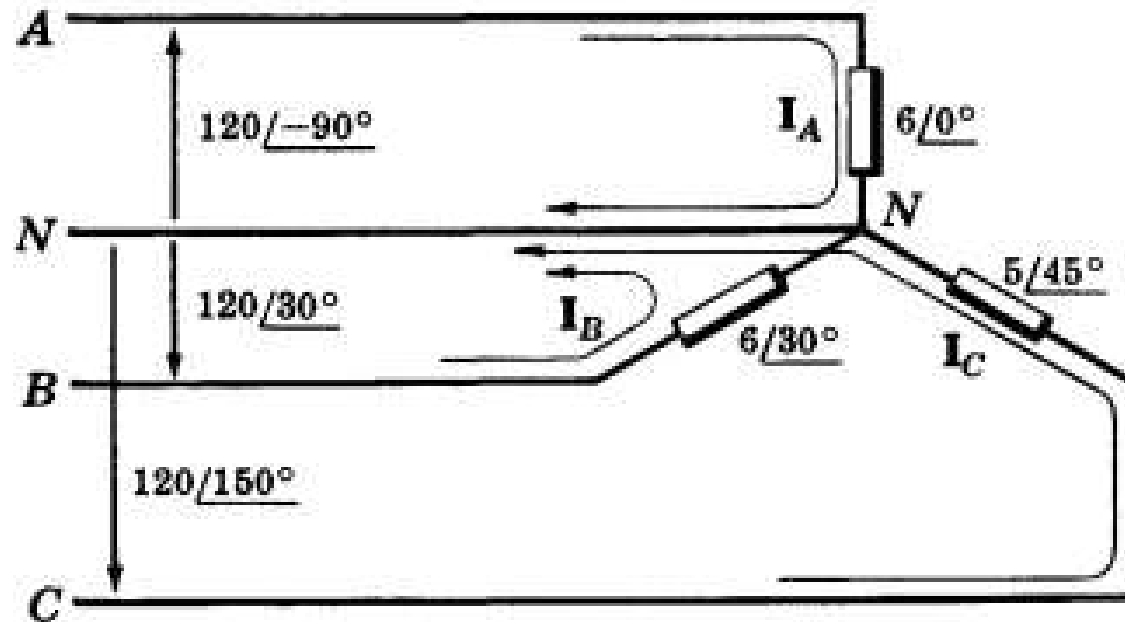
A three-phase, four-wire, 208 volt, CBA system has a wye-connected load with

$$Z_A = 6\angle 0^\circ,$$

$$Z_B = 6\angle 30^\circ, \text{ and}$$

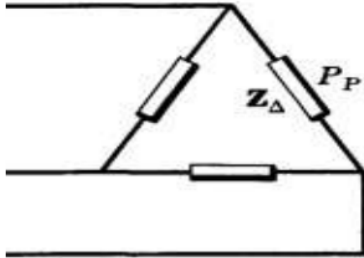
$$Z_C = 5\angle 45^\circ.$$

Obtain the three line currents and the neutral current. Draw the phasor diagram.



Power in balanced three-phase loads

Since the phase impedances of balanced wye or delta loads contain equal currents, the phase power is one-third of the total power.



- The voltage across is line voltage
- The current is phase current.
- The angle between V & I is the angle on the impedance.

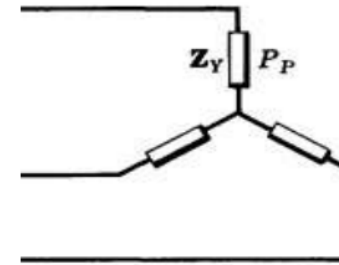
Phase power $P_P = V_L I_P \cos \theta$

Total power $P_T = 3 V_L I_P \cos \theta$

For a balanced Δ -connected loads:

$$I_L = \sqrt{3} I_P$$

$$P_T = \sqrt{3} V_L I_L \cos \theta$$



- The voltage across is phase voltage
- The current is line current.
- The angle between V & I is the angle on the impedance.

Phase power $P_P = V_P I_L \cos \theta$

Total power $P_T = 3 V_P I_L \cos \theta$

For a balanced Y-connected loads:

$$V_L = \sqrt{3} V_P$$

$$P_T = \sqrt{3} V_L I_L \cos \theta$$

Power in balanced three-phase loads

$$P_T = \sqrt{3}V_L I_L \cos \theta$$

$$Q_T = \sqrt{3}V_L I_L \sin \theta$$

$$S_T = \sqrt{3}V_L I_L$$